Enron Corp.

Risk Assessment & Control

Presented by:

Ted Murphy
Vice President

Enron Corp. - Business Units

Transportation & Distribution

Gas Pipeline Group

Portland General Electric

Wholesale Energy Operations and Services

Enron Capital & Trade
- North America
- Europe

Enron International

Retail Energy Services

Enron Energy Services

New Businesses

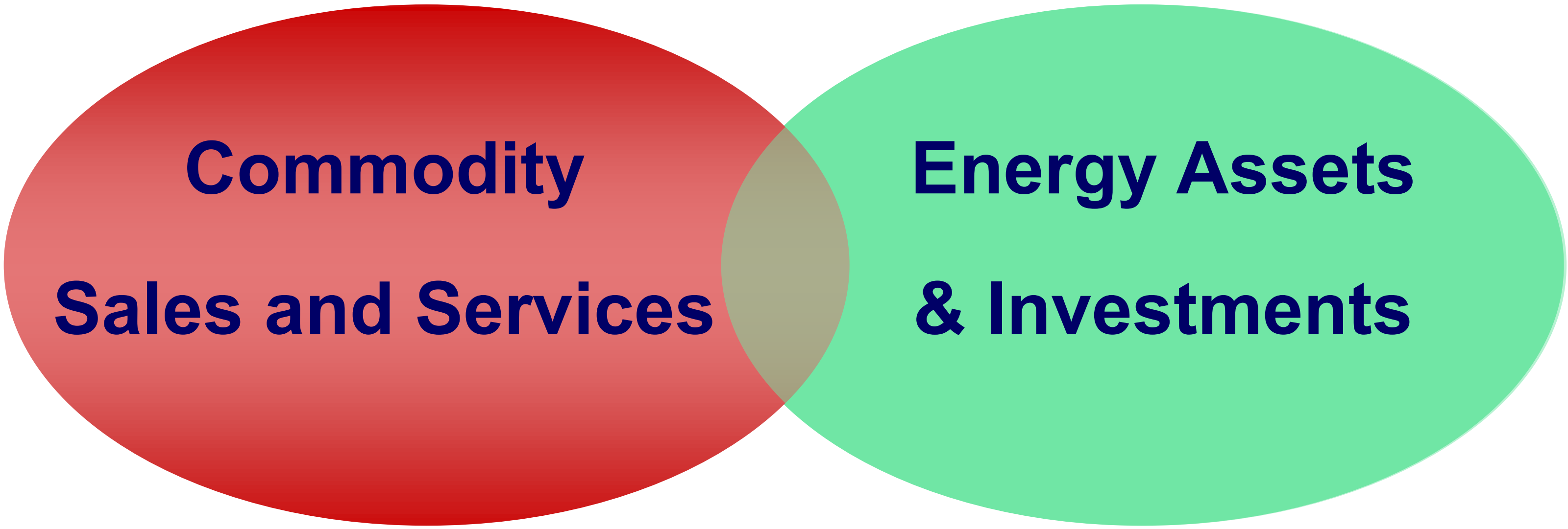
Enron Communications

Azurix

Exploration & Production

Enron Oil & Gas Company

Key Elements of Enron's Wholesale Business



Worldwide Marketing and Sales of Energy Commodities

Price Risk Management Products

Development, Construction and Operation of Energy Assets Worldwide

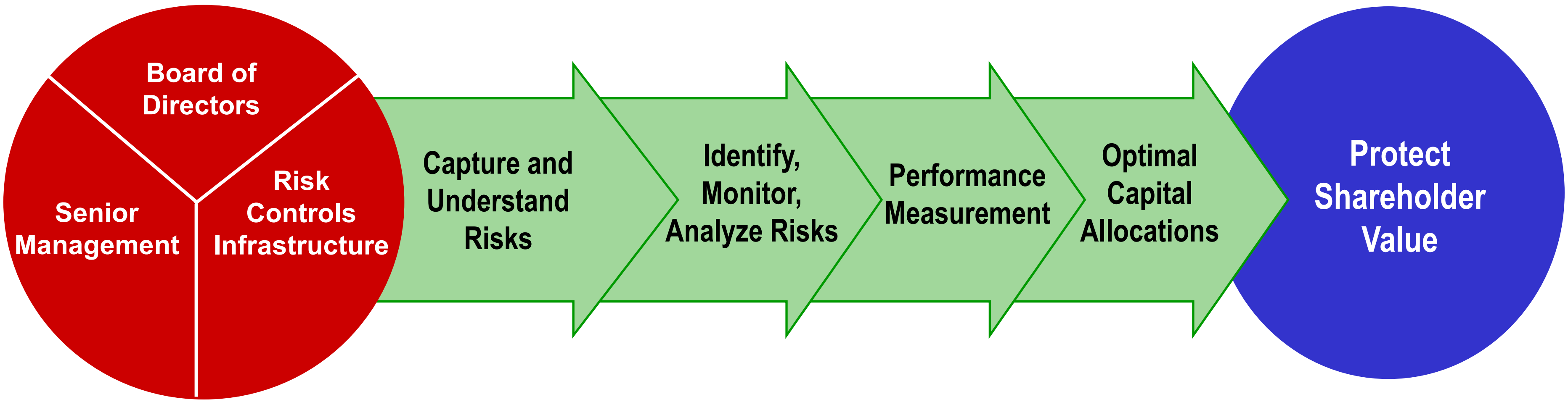
Origination and Management of Large Energy Investment and Finance Portfolio

Risk Assessment & Control Group ("RAC")

126 Professionals

- **Enron Board of Directors**
  - **President & COO Enron** — *Jeffrey K. Skilling*
    - **Chief Risk Officer** — *Richard B. Buy*
      - **Market Risk Management Controls** — *Ted Murphy*
      - **Underwriting** — *John S. Hopley*
      - **Risk Analytics (RAROC)** — *Mark A. Ruane*
      - **Resource Evaluation** — *Monte L. Gleason*
      - **Credit Risk Management** — *William S. Bradford*
      - **Research** — *Vince Kaminski*

Evaluating Risk



Underwriting

Products

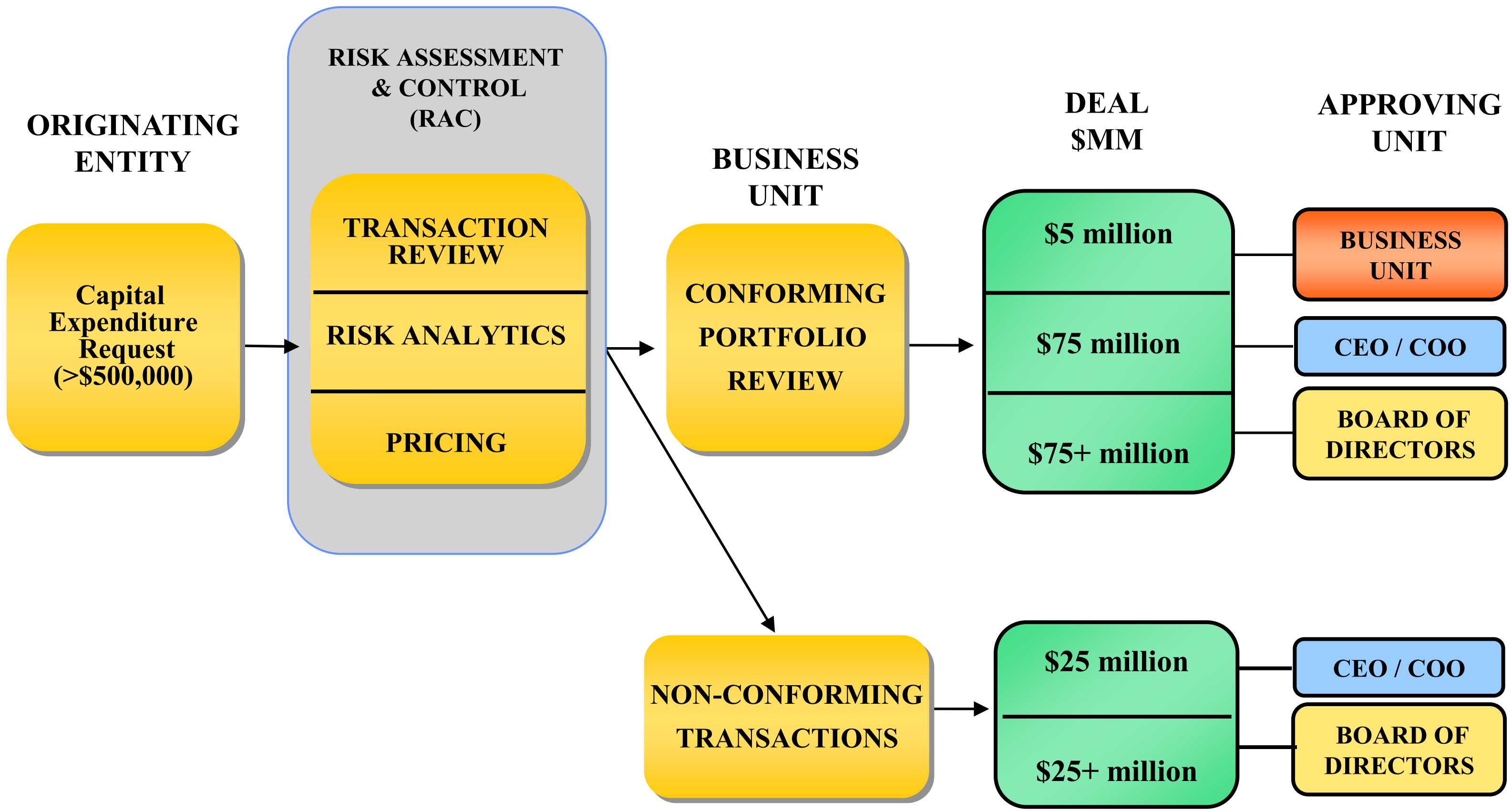
- Project Finance
- Pre-Paid Forward Contracts
- Subordinated Debt
- Warrants
- Corporate Equity

Domestic and International Markets

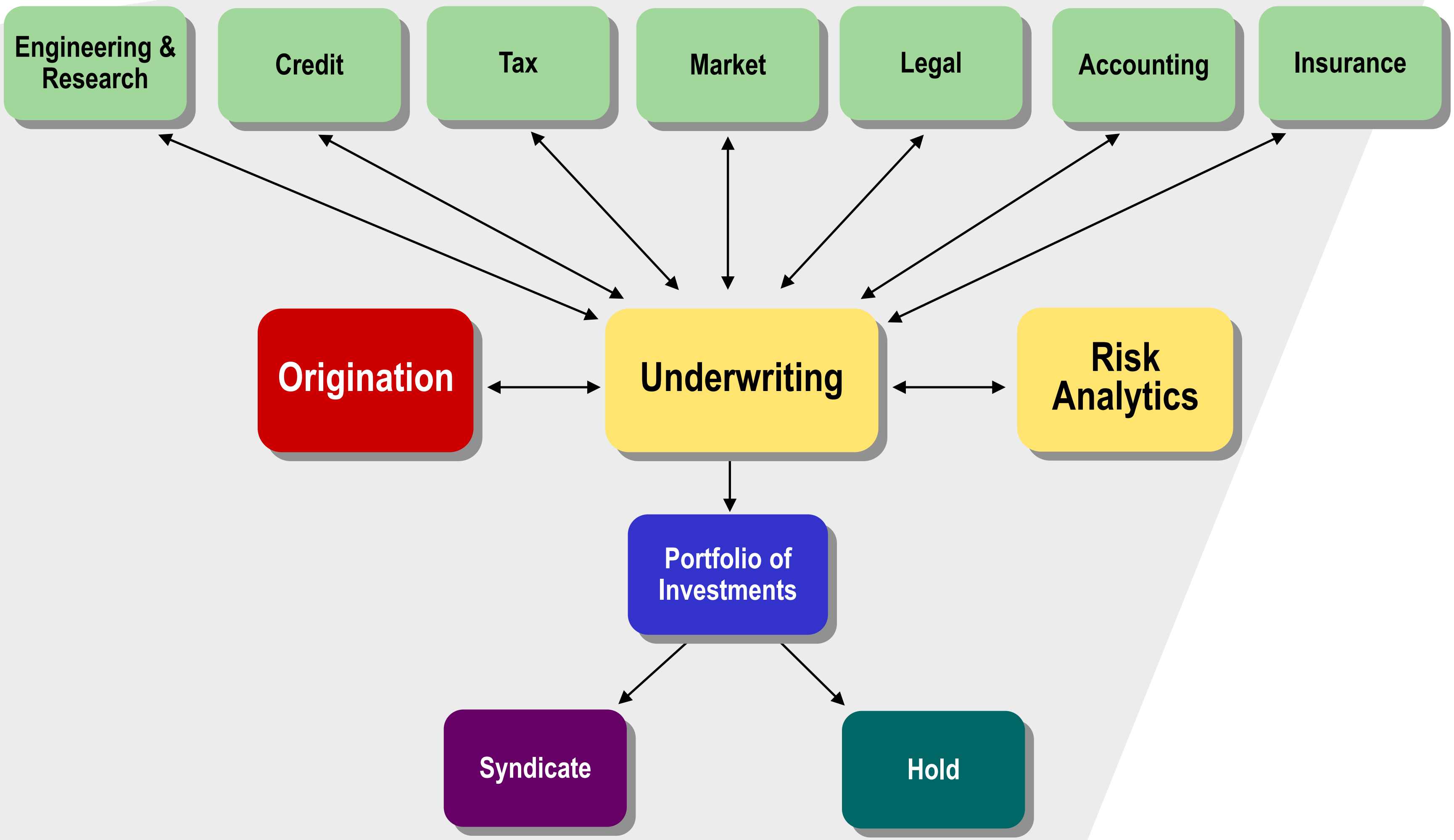
- Gas
- Power
- Oil
- Liquids
- Coal
- Metals
- Paper
- Steel
- Telecom
- Pipelines
- Oilfield Services

Transaction Approval Process

Enron Board of Directors Actively Reviews Capital Transactions



Transaction Review



Risk Analytics Process

TRANSACTION REVIEW

RISK ANALYTICS

PRICING

Deterministic Transaction Model

- Review Models and Methodologies
- Analyze Risk and Assumptions

Probabilistic Transaction Model

- Apply Distribution to Capture Risk
- Simulate Possible Outcomes

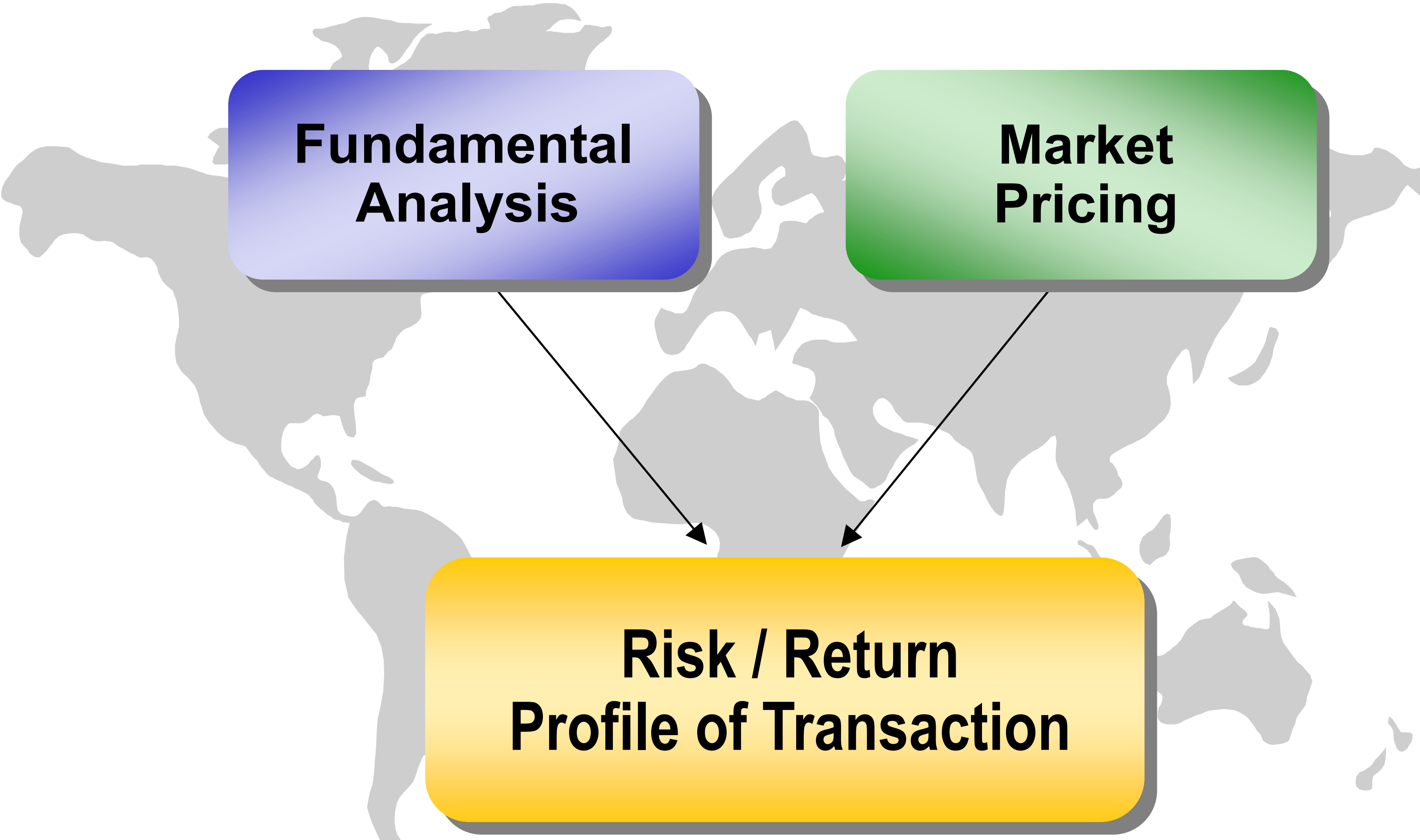
Transaction Approval

- Multiple Sign-Offs Required
- Requires Completion of Risk Analytics

Ongoing Evaluation

- Quarterly Review of Transaction Performance
- Senior Management Reporting

Risk Assessment Process



Example Transaction Approval Document

ENRON RISK ASSESSMENT AND CONTROL
DEAL APPROVAL SHEET

DEAL NAME:	EXAMPLE TRANSACTION	**Date DASH Completed:**	12/18/98
Counterparty:		**RAC Analyst:**	
Business Unit:	Azurix	**Investment Type:**	
Business Unit Originator:	J. Paul Oxer	**Capital Funding Source(s):**	
☐ **Public** ☒ **Private**		**Expected Closing Date:**	
☐ **Merchant** ☒ **Strategic**		**Expected Funding Date:**	
☒ **Conforming** ☐ **Nonconforming**		**Board Approval: X Pending** ☐ **Received** ☐ ☐ **N/A**	

DEAL DESCRIPTION

Strategic Interest is a water and sewage company in a South American Country (SAC). Acquire an interest from the SAC government of between 39.8% and 50% of the Strategic Interest, and through a Shareholders' Agreement be granted control over all but a limited number of corporate matters. It serves a population of 2.0 million and has approximately 400,000 water customers and 300,000 wastewater customers. All customers are metered with a collection rate at approximately 95%. Average water losses of 40% can be reduced to 30% in five years. Azurix requests approval to acquire the Strategic Interest for cash consideration of up to $150 million.

TRANSACTION SOURCES AND USES OF FUNDS ($ Thousands)

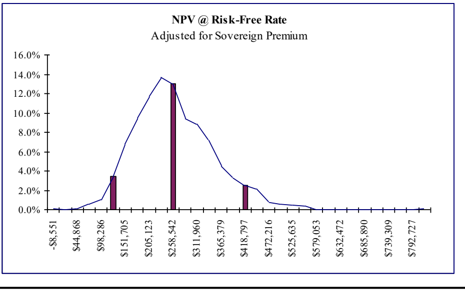
	Sources		Uses
Enron Equity	$140,000	Acquisition	$140,000
Total	$140,000		$140,000

SUMMARY (note – each instrument must be shown – options must be shown separately)

	($000)		
Capital Commitment:	$140,000	Weighted Average Life (yrs.):	19.00
Bid Bond Amount:	**$10,000**		

Return Components:	PV @ Capital Price	Cumulative IRR	Capital Price Components	
Cash Outflows	148,737	N/A	Risk free rate (%):	5.50
Fees	0	0.00%	Equity/Credit premium (%):	5.00
Intermed. Cash Flows	140,014	11.95%	Country Premium (%):	3.50
Terminal Value	54,983	18.50%	Other (%):	3.00
Total NPV	**66,169**	**18.50%**	**RAC CAPITAL PRICE:**	**17.00%**

EXISTING EXPOSURE



NON-HEDGABLE (IDIOSYNCRATIC) RISKS

RISK	DESCRIPTION	MITIGATION/COMMENTS
Construction Cost Overruns and Delays		
Management Risk		
Heat Rate Risk		
Operator Risk		
Operating Expense Risk		
Trapped Cash Risk		
Bid Bond Risk		
Equity Syndication Risk		
Country Risk		

HEDGABLE OR MARKET RISKS

Foreign Exchange Risk	Include curves used, description of hedges in place, functional currency of the transaction..
Interest Rate Risk	Include interest rate basis of outstanding or projected debt. Describe hedges in place at transaction level. Does not include issues related to Enron's funding of the transaction. Note that all transactions carry interest rate exposure through the discount rate.
Equity Risk	Include for all non-structured credit transactions. Include discussion of adjustments to value made for lack of liquidity. Note that equity market risk is reflected as a component both of the exit multiple and the discount rate. Market comparables are required for all private merchant transactions. List comps and the associated multiples used if applicable.
Credit Risk	Include for all structured credits. Describe rating (implied or otherwise) for the counterparty. Include discussion of how credit risk is modeled. Note that credit risk is reflected as a component of the discount rate.
Inflation Risk	Discuss impact of inflation on cash flows
Commodity Risk	Include commodity volumes, base curves, source of curves, relative market liquidity, hedges in place at transaction level. Include attachments if necessary.

SYNDICATION (ECM):

- ☐ Immediately syndicatable at current capital price
- ☐ Syndication within one year at current capital price
- ☐ Not syndicatable at current capital price
- ☐ N/A

APPROVALS:	Name	Signature	Date
RAC Management	John Hopley	John Hopley	1-21-99
Enron Capital Management	Andy Fastow/Jeff McMahon	Andy Fastow	1-21-99
Business Unit Originator	J. Paul Oxer	J P Oxer	1-21-99
Business Unit Legal	Sheila Tweed	Sheila Tweed	1-21-99
Portfolio Manager	Jere Overdyke/Cliff Shedd	Jere Overdyke	1-21-99
Business Unit Mgmt	Joe Sutton	Joe Sutton	1-21-99
ENE Management	Jeffrey Skilling	Jeff Skilling	1-21-99

Underwriting Summary

Assurance that returns are evaluated relative to the risks assumed

Provide independent quantitative and qualitative feedback to senior management.

Mitigate risks in transactions where appropriate.

Coordinate other internal resources (tax, legal, accounting, insurance, trading, etc.) for transaction evaluation and capital allocation.

Participate in negotiations with origination

Assist in syndication efforts

Transaction Universe

Credit Risk Management

Physical and Financial Products

- Forwards
- Swaps
- Options
- Swaptions
- Exotics

Over $4 billion in price risk management assets

Over $1 billion in A/R exposure

Domestic and International Markets

- Gas
- Power
- Oil
- Liquids
- Chemicals
- Coal
- Weather
- Paper
- Equities
- Interest Rate
- Currency

Over 80% of exposure to investment grade counterparties

8,000 counterparties, 2,500 active

Group Of Thirty

Credit Risk Measurement and Management Recommendations from the Publication Derivatives: Practices and Principles

- 10 Measuring Exposure
- 11 Aggregating Exposure
- 12 Independent Credit Risk Management
- 13 Master Agreements
- 14 Credit Enhancement

The Group of Thirty is a private, independent, body established in 1978 to provide analysis and recommendations for the financial markets

Credit Risk Management Drivers

Counterparty Credit Evaluation — **Counterparty Credit Exposure**

Management Tools

- Information Systems
- Credit Reserve
- Contract Terms
- Credit Enhancement
- Credit Scoring: E-Rating

Counterparty Credit Evaluation

Assess Counterparty's Ability and Willingness to Honor Obligations

Quantitative Analysis	Qualitative Analysis
Financial strength	**Industry sector**
Key financial ratios	**Rank within industry**
• Operating performance	**Management**
• Liquidity	**Trading activities and controls**
• Debt service	**Sovereign risk**
• Leverage	

Establish Internal "Enron Rating" for All Counterparties

Credit Reserve

Credit priced at transaction level via Monte Carlo simulation

<u>Factors in simulation model</u>

- International Counterparty Credit Rating
- Probability of default derived from corporate bond spreads
- Tenor of deal
- Price Volatility
- Deal Pricing vs. Current Price Curve
- Interest Rates
- Credit Enhancements

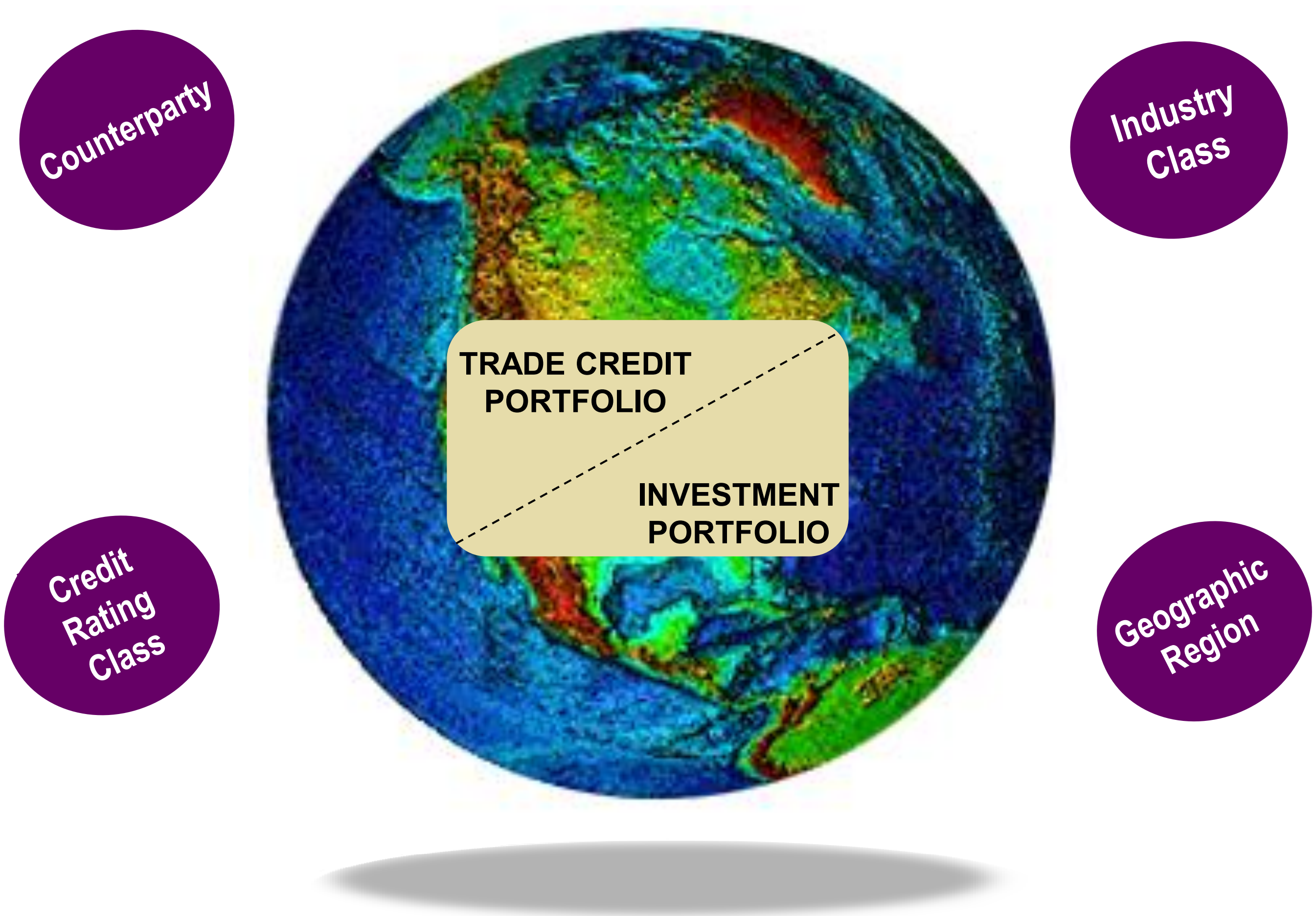
Reserve held against MTM earnings reflects market price of credit on transaction

Credit reserve valued on portfolio basis monthly to confirm adequacy

Credit Risk Management Summary

Manage global credit exposure

Monitor and limit concentration risks



Responsibilities of Risk Management Controls

Administration / Guardianship of the Enron Corp. Risk Management Policy

Commodity Portfolio Risk Analysis

Limits are reviewed daily by Risk Management Controls

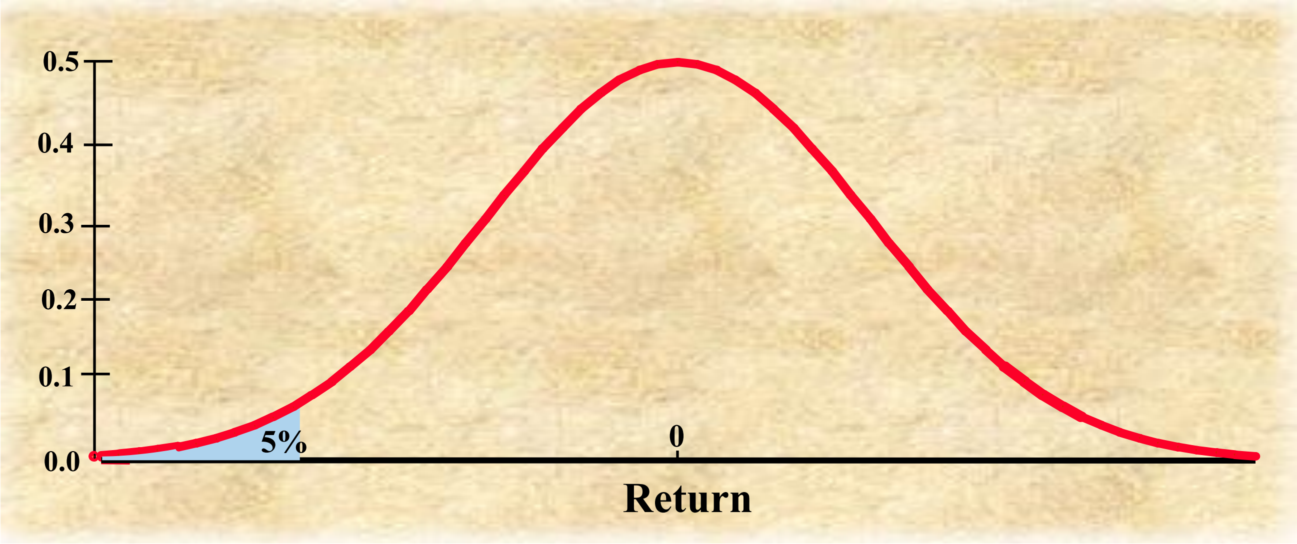
Limits are reviewed and ratified by the Enron Board of Directors

Value-at-Risk

Value-at-Risk (V@R) has become the standard measure of risk across portfolios and businesses

- Dollar loss that may be experienced in the value of a portfolio
  - » over a defined time period
  - » with a given confidence interval
  - » due to changes in market risk factors

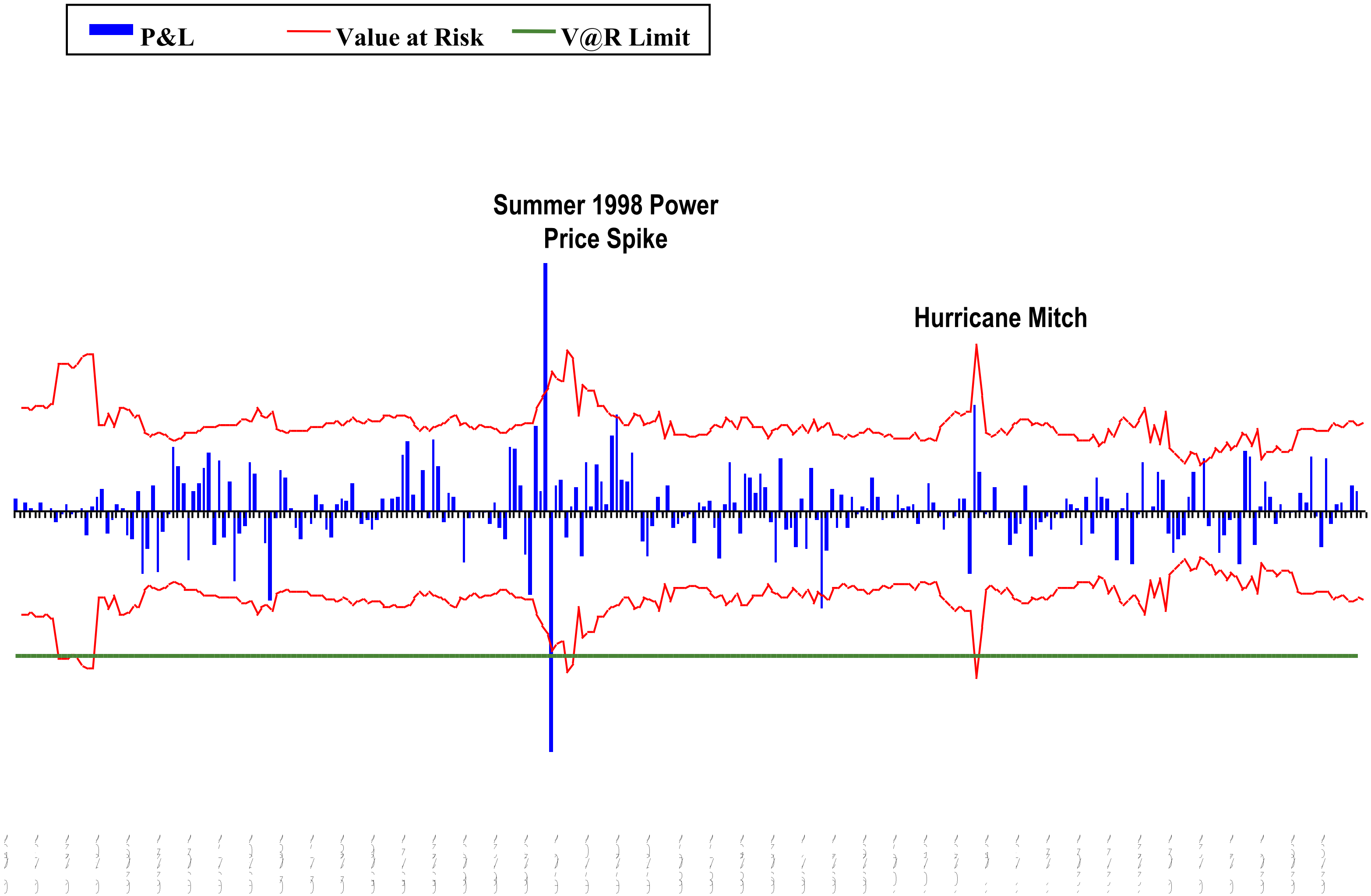
Enron Corp.'s V@R methodology uses a 95% confidence interval and a one-day time horizon



Other Measures

- Stress testing
- Scenario analysis

Backtesting ENRON V@R - 1/1/98 to 2/26/99



Checks and Balances

Risk Control personnel on the trading floor

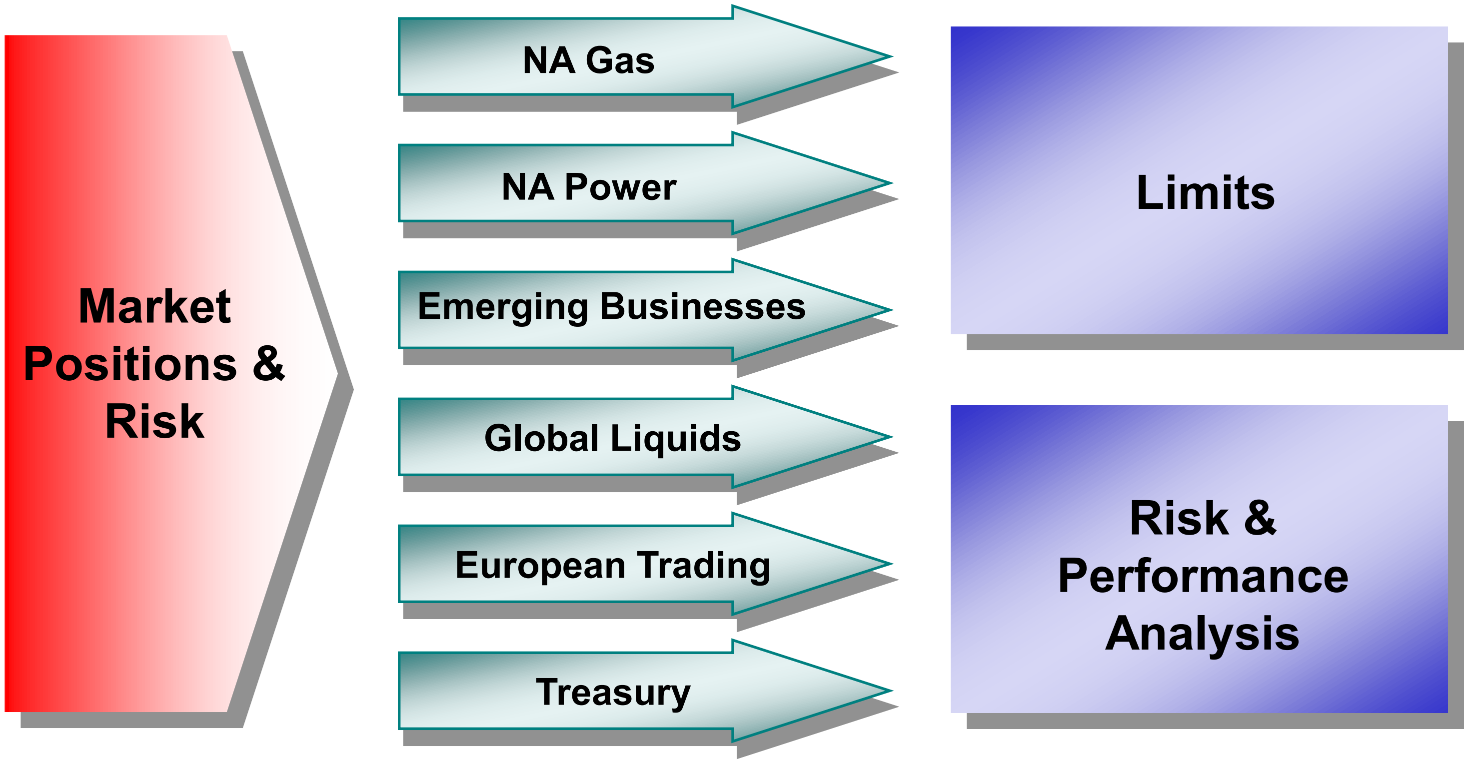
Multi-disciplined assessment team

Expertise of staff

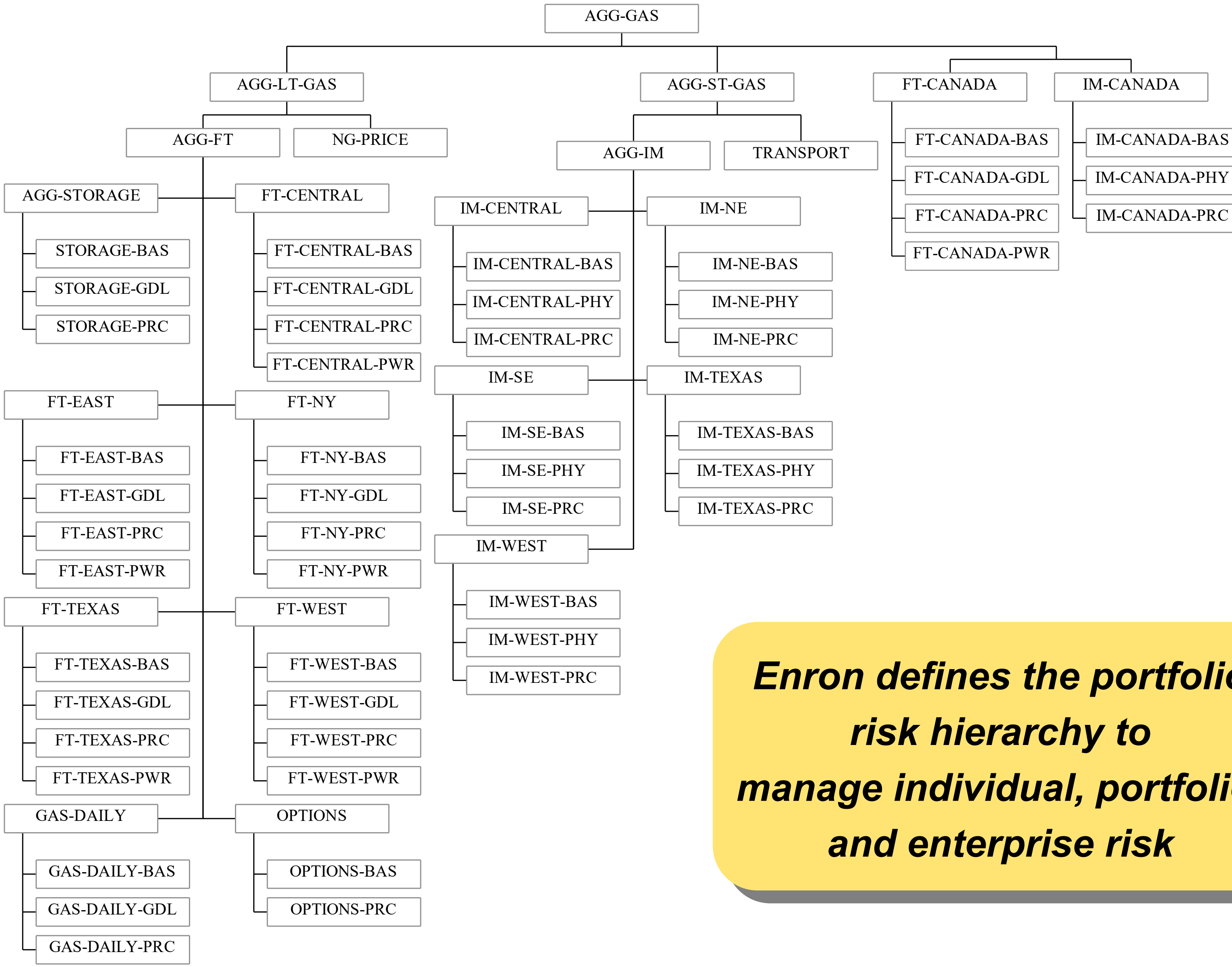
Senior Management Review

Risk Management Controls - Major Portfolios

PORTFOLIOS



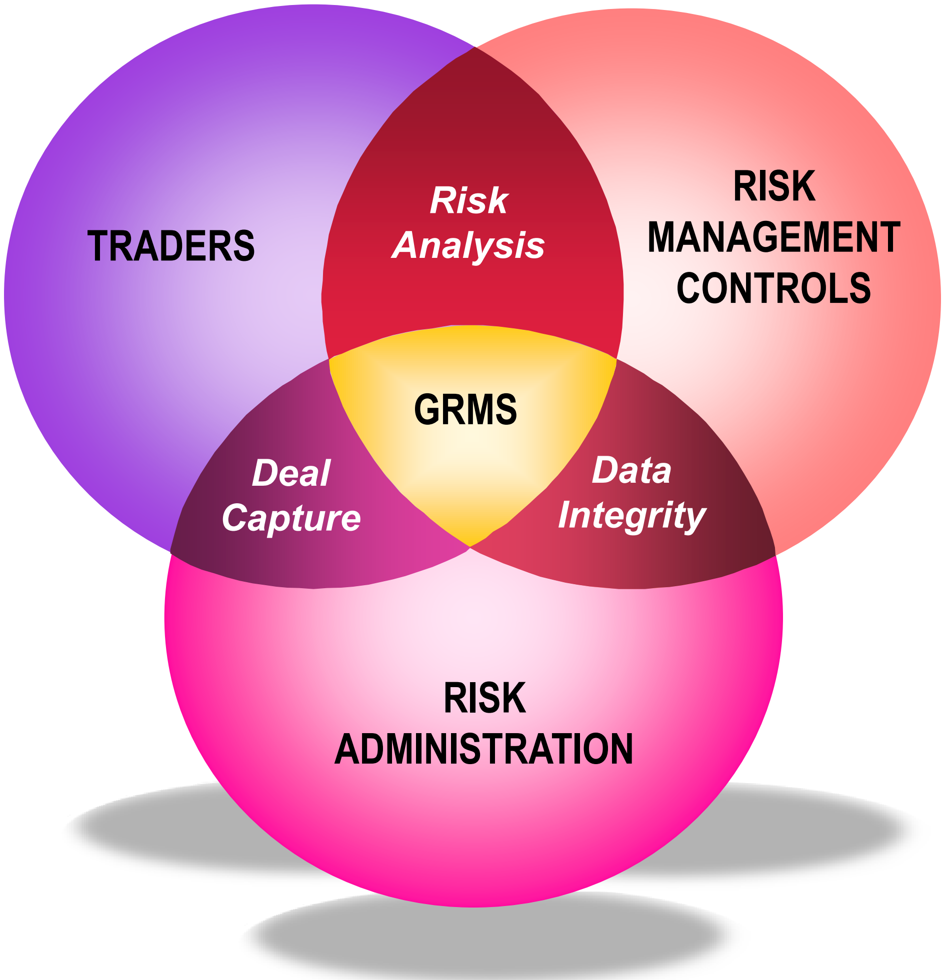
ECT North America - Natural Gas Risk Book Hierarchy



Enron defines the portfolio risk hierarchy to manage individual, portfolio, and enterprise risk

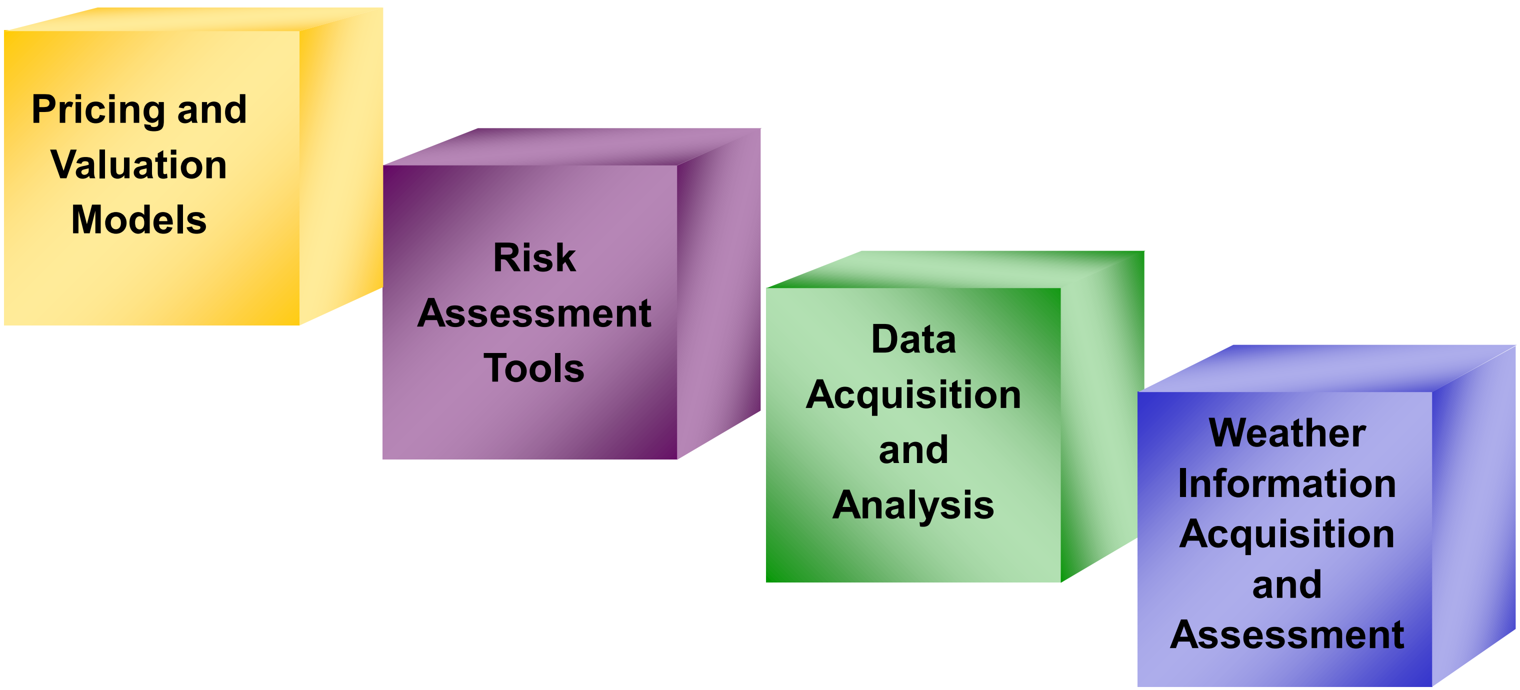
Global System Integration - Risk Management Controls

The Global Risk Monitoring System (GRMS) is a tool for evaluating and analyzing the risks associated with Enron's trading positions globally.



Research Group Main Functions

Centralized development of accurate and consistent quantitative models across Enron



Strong Educational Background

7 Ph.D.s In Physics

1 Ph.D in Computer Science

2 Ph. D. In Economics

1 Ph.D. In Operations/Research 1

1 Ph.D. Earth Sciences/ J.D.

1 Ph.D. Mathematics

1 Ph.D. In Material Science

1 Ph.D.s In Engineering

1 Ph.D. Electrical Engineering

1 Ph.D. Earth Science

1 MBA/ M.S. Physics

1 M.S. Mathematics

1 M.S. Economics

1 M.S. Business

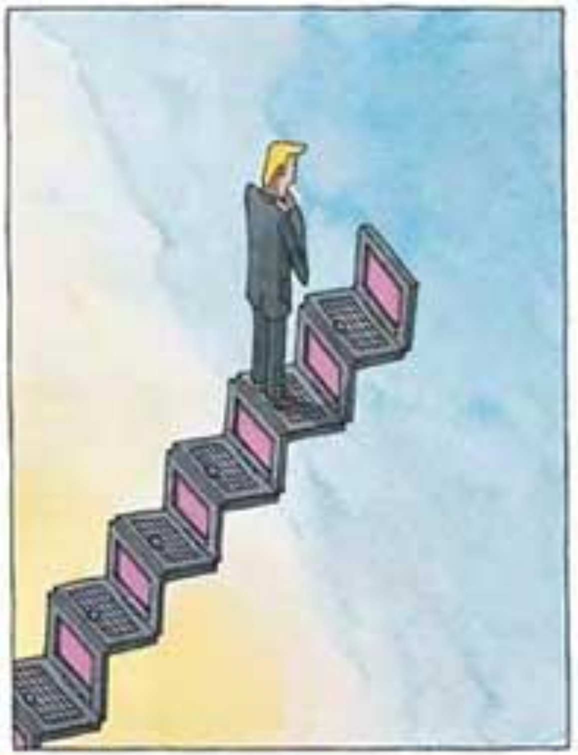
2 MBAs

2 Student Candidates for MBA

Staying on the Cutting Edge of Technology

Monitoring Important Developments:

- Financial Economics
- Applied Mathematics
- Other Fields of Interest to Enron



Conclusion

Enron has invested in the people, tools, and systems necessary for effective risk management.

Enron highly values an independent group focused on controls.

Integration of multiple disciplines in Risk Management Controls Group enhances overall effort.

The Board of Directors' involvement indicative of Enron's focus and commitment to monitor risk.

Commitment includes applying sophisticated concepts (relative to financial securities industry) that are appropriate for Enron's unique markets and risks.